FESTRONIX

Website Product Requirements Document

Immersive 3D public website + event registration and operations platform

Document field	Details
Product	Official Festronix Website & Management Platform
Institution	Vardhaman College of Engineering, Hyderabad
Status	Draft —
Version	1.0
Prepared	12 October 2026
Target launch	Within 1–2 weeks, subject to content and approvals
Expected registrations	Approximately 2,000–5,000

Purpose Define the product, users, workflows, scope, design direction, acceptance criteria and technical approach. Unsettled choices are explicitly marked as open.

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1. Executive Summary

Festronix is the annual technical fest of Vardhaman College of Engineering, Hyderabad. The product combines a futuristic, immersive 3D promotional website with a practical event registration and management platform.

SAC is the central organising body. Each department has a Chief Coordinator, and each individual event has an Event Coordinator. Event Coordinators manage their own events. Participants may come from Vardhaman or other colleges and can participate individually, as a duo, or as a team.

The platform supports event discovery, online registrations, configurable capacity, UPI/QR payment instructions, payment screenshot and transaction-ID submission, manual proof review, spot registrations entered by Event Coordinators, digital passes, QR check-in, certificates, announcements and reports.

> Product principle Make the public experience memorable and cinematic while keeping registration, payment review and event-day operations fast, accessible and reliable.

2. Product Vision and Goals

2.1 Vision

Create the official digital home of Festronix: an immersive technology-fest experience backed by one reliable system for event discovery, registrations, teams, payment evidence and event operations.

2.2 Goals

- Deliver a premium, futuristic website with carefully selected 3D moments.
- Let participants discover events across departments.
- Support individual, duo and team participation with event-specific rules.
- Provide one source of truth for online and spot registrations.
- Allow Event Coordinators to manage assigned events.
- Track UPI/QR payment evidence with screenshot and transaction ID/UTR, followed by human verification.
- Support QR passes, check-in, certificates, announcements and CSV/Excel reports.
- Support an expected 2,000–5,000 registrations, subject to testing.

2.3 Success measures

Measure	Definition
Registration	Participants can complete valid registrations without unnecessary friction.
Coordinator visibility	Coordinators can see registrations and payment review status for their assigned events.
Spot registration	Staff can create solo, duo and team registrations on fest day.
Payment evidence	Proof is stored securely and remains pending until reviewed.
Check-in	Unique QR tickets are validated and duplicate check-ins are detected.
Performance	Core discovery and registration flows remain usable on mobile and typical campus networks.

3. Users and Organisational Structure

Role	Responsibilities	Access boundary
Public visitor	Browse fest information, events, prizes, sponsors and rules.	Public content only.
Participant / team leader	Profile, registrations, teammates, payment proof, passes and certificates.	Own profile and registrations; permitted team details.
Event Coordinator	Manage assigned event details, registrations, spot registrations, payment evidence, check-in and reports.	Assigned event(s) only unless separately authorised.
Department Chief Coordinator	Monitor department events, capacity, registrations and readiness.	Their department's events.
SAC / super-admin	Oversee fest, departments, assignments, accounts, roles, content and reports.	Fest-wide administration.
Finance / payment reviewer	Review payment evidence and collection reports.	Payment data needed for assigned duties.
Volunteer / check-in staff	Scan tickets and support entry operations.	Limited check-in access.

The final role names and permission matrix require SAC approval. Permissions must be enforced on the server/database, not only by hiding UI controls.

4. Scope and Priorities

4.1 MVP — required for launch

- Responsive public website with 3D-led homepage, event directory and event detail pages.
- Authentication and participant registration.
- Individual, duo and team formats with configurable team-size rules.
- Event-specific fees, deadlines and capacity limits.
- Event Coordinator dashboard for assigned events.
- Online registrations and coordinator-entered spot registrations.
- UPI/QR payment instructions plus screenshot and transaction-ID/UTR submission.
- Manual payment review with pending, verified and rejected states.
- Capacity checks and duplicate prevention.
- Digital pass / QR generation and check-in validation.
- Basic announcements, CSV/Excel export and role-based access.
- Database security policies and audit records for important actions.

4.2 Next priority, if time permits

- Certificate uploads and participant downloads.
- Department Chief Coordinator dashboard and consolidated SAC analytics.
- Advanced filters, attendance summaries and event-readiness indicators.
- Sponsor and content management through an admin interface.
- Additional interactive 3D scenes and polish.

4.3 Exclusions unless separately approved

- Automatic payment gateway confirmation or automatic refunds.
- Native iOS or Android application.
- Complex offline-first synchronisation.
- Unrestricted custom CMS or custom payment processor.
- 3D effects that block registration or make the website unusable on mobile.

5. Sitemap

Area	Proposed routes / screens
Public	/, /events, /events/[slug], /departments, /prizes, /sponsors, /about, /guidelines, /faq, /contact, /privacy, /terms
Authentication	/login, /register, /forgot-password
Participant	/student, /student/registrations, /student/registrations/[id], /student/teams, /student/passes, /student/certificates, /student/profile
Event Coordinator	/admin/events/[eventId], /registrations, /spot-registration, /payments, /check-in, /reports (scoped to assigned event)
Department Coordinator	/admin/departments/[departmentId], departmental events and reports
SAC / super-admin	/admin, /admin/departments, /admin/events, /admin/users, /admin/roles, /admin/content, /admin/announcements, /admin/sponsors, /admin/reports, /admin/settings
Finance	/admin/payments, /admin/collections, /admin/reports/payments

These routes are a proposed information architecture. They do not all need to appear as separate top-level navigation items.

6. Functional Requirements

6.1 Public website and discovery

- FR-PUB-01: Homepage introduces Festronix and links clearly to events and registration.
- FR-PUB-02: Event directory supports search and filters for department, category, participation format and status.
- FR-PUB-03: Event pages show description, rules, eligibility, format, team size, fee, capacity, deadline, date/time, venue, prizes and contact details when configured.
- FR-PUB-04: Event state is clear: upcoming, open, full, closed or completed.
- FR-PUB-05: Routine public content should be editable by authorised staff where admin content tools are in scope.
- FR-PUB-06: Categories should accommodate technical events, workshops, non-technical events, hackathons/project competitions, gaming/esports and department-specific events.

6.2 Accounts and profiles

- FR-AUTH-01: Participants can sign up, sign in, sign out and recover access.
- FR-AUTH-02: Profile captures only necessary information, such as name, email, phone, college and required student details.
- FR-AUTH-03: The system distinguishes registration creator/team leader from added teammates.
- FR-AUTH-04: Participants can view registration and payment status, event details, passes and available certificates.
- FR-AUTH-05: Personal information is not exposed publicly without an approved purpose.

6.3 Event configuration

- FR-EVT-01: Each event belongs to a department and has an assigned Event Coordinator.
- FR-EVT-02: Event Coordinators manage details and registrations for assigned events.
- FR-EVT-03: Each event configures individual, duo or team participation.
- FR-EVT-04: Duo/team events can configure minimum and maximum team size.
- FR-EVT-05: Each event can be free or paid; fee and payment instructions are configurable.
- FR-EVT-06: Each event can set its own deadline and capacity.
- FR-EVT-07: Event details include schedule, venue, rules, FAQs and contacts.
- FR-EVT-08: Direct publishing vs SAC approval must follow the policy SAC chooses; this is not yet confirmed.

6.4 Registrations and teams

- FR-REG-01: Participants can register when eligible and the event is open.
- FR-REG-02: One registration creator can add all teammates for duo/team participation.
- FR-REG-03: Required team size and teammate details are validated.
- FR-REG-04: Every registration has a unique ID and source (online or spot).
- FR-REG-05: Duplicate registration prevention follows an approved event rule.
- FR-REG-06: Capacity checks must avoid overbooking during concurrent registrations.
- FR-REG-07: Registration status and payment status are tracked separately.

6.5 Payment proof and review

- FR-PAY-01: Free events do not require payment proof.
- FR-PAY-02: Paid events show the configured UPI/QR payment instructions.
- FR-PAY-03: Participants submit both payment screenshot and transaction ID/UTR reference.
- FR-PAY-04: Proof files are private and visible only to authorised reviewers.
- FR-PAY-05: Submission sets payment to Verification pending; it never automatically marks payment as verified.
- FR-PAY-06: An authorised reviewer can verify or reject proof and provide a rejection reason.
- FR-PAY-07: Reviewer identity and review timestamp are recorded.
- FR-PAY-08: Collection account/provider must be approved before live launch.
- FR-PAY-09: MVP uses manual proof review; automatic gateway confirmation is excluded unless separately approved.

6.6 Spot registration

- FR-SPOT-01: An Event Coordinator can create registrations at the venue.
- FR-SPOT-02: Spot registration supports individual, duo and team formats.
- FR-SPOT-03: It captures the same required participant/team information as online registration.
- FR-SPOT-04: Eligibility, duplicates and capacity are checked.
- FR-SPOT-05: Source is recorded as spot, with the staff member who entered it.
- FR-SPOT-06: Applicable fee and payment status are recorded according to the authorised process.
- FR-SPOT-07: A pass/QR identifier can be generated according to the event's ticket policy.

6.7 QR tickets and check-in

- FR-QR-01: Each eligible registration receives a unique QR identifier.
- FR-QR-02: Staff can scan a QR or search a registration manually.
- FR-QR-03: Results distinguish valid, invalid, already checked-in and not eligible.
- FR-QR-04: Check-in is timestamped and attributed to staff where possible.
- FR-QR-05: QR codes do not expose sensitive participant data directly.

6.8 Announcements, certificates and content

- FR-CONT-01: Authorised staff can publish fest-wide, department-specific or event-specific announcements.

- FR-CONT-02: Authorised staff can upload certificates and make them available to eligible participants.
- FR-CONT-03: Sponsors, prize details, FAQs and general content should be editable through admin tools where feasible.

6.9 Reports and exports

- FR-REP-01: Event Coordinators see registration counts and participant lists for assigned events.
- FR-REP-02: SAC sees consolidated registration and event summaries.
- FR-REP-03: Authorised users can export registration data to CSV or Excel.
- FR-REP-04: Payment reports separate expected fees, pending proof, verified and rejected submissions; unverified proof must not be reported as confirmed collection.
- FR-REP-05: Exports respect permissions and minimise unnecessary personal data.

7. Registration and Payment Workflows

7.1 Online registration

1. Participant opens an event page and reviews eligibility, format, capacity, deadline and fee.
2. Participant signs in or creates an account.
3. Participant selects individual, duo or team participation.
4. For duo/team events, the registration creator enters teammate information.
5. System validates eligibility, team size, deadline, duplicate rules and capacity.
6. Free event: registration is recorded without payment.
7. Paid event: participant pays using the configured UPI/QR instructions and submits screenshot plus transaction ID/UTR.
8. Registration and payment states appear separately in the participant dashboard.
9. Authorised reviewer checks evidence and marks it verified or rejected.
10. Participant can view status and access the pass according to the event policy.

7.2 Spot registration

11. Coordinator signs in and selects an assigned event.
12. Coordinator opens spot registration.
13. Coordinator selects individual, duo or team and enters participant details.
14. System checks eligibility, duplicate rules and available capacity.
15. Coordinator records applicable fee and payment status under the approved process.
16. System saves source = spot and records who entered the registration.
17. System generates a pass/QR identifier if allowed by the event policy.

7.3 Payment states

State	Meaning	Next action
Payment required	No proof submitted for a paid event.	Participant submits screenshot and reference.
Verification pending	Proof submitted but not reviewed.	Reviewer verifies or rejects.
Verified	Authorised reviewer accepted evidence.	Retain audit record and update status.
Rejected	Evidence not accepted.	Provide reason; allow resubmission if policy permits.
Not applicable	Event is free.	No payment review needed.

Important A screenshot or transaction ID alone does not independently prove settlement. The platform records evidence and an authorised human decision; it does not claim automatic bank confirmation.

8. Admin and Coordinator Workflows

8.1 Event Coordinator

- See assigned events by default.
- Edit event details, rules, team limits, fee, capacity, deadline, schedule and venue, subject to permissions.
- Review registrations and payment evidence.
- Create spot registrations on fest day.
- Validate QR tickets and view attendance.
- Publish event announcements and export event data.

8.2 Department Chief Coordinator

- View events within the department.
- Monitor registrations, capacity, payment review queues and readiness.
- Access departmental reports and coordinate with Event Coordinators.

8.3 SAC / super-admin

- Manage departments, categories, event assignments, accounts, roles and permissions.
- Oversee events, registrations, content, sponsors, announcements and reports.
- Access fest-wide analytics and consolidated exports.
- Set fest-level settings and decide the event publication policy.

8.4 Finance / payment reviewer

- Access payment information required for assigned duties.
- Review screenshots and transaction references.
- Verify/reject proof and record decisions.
- Export payment/collection reports when authorised.

9. 3D Design and Experience Requirements

The visual direction is a futuristic technology festival: dark base, neon accents, immersive 3D objects and cinematic transitions. The design must feel premium without sacrificing readability, accessibility or registration speed.

9.1 Visual direction

- Dark backgrounds with controlled neon cyan and violet accents.
- Cinematic 3D hero scene using abstract digital structures, holographic surfaces, particles, light trails or a futuristic arena motif.
- Prominent Festronix title, clear primary action and readable navigation.
- Interactive depth and motion on selected event cards and sections.
- Consistent design tokens for colour, typography, spacing, borders, shadows and motion.

9.2 Motion and implementation

- Use scroll-linked transitions only when they improve the story or orientation.

- Keep registration forms and admin tables restrained and functional.
- Respect reduced-motion preferences; provide a static/fallback experience.
- Avoid effects that consume excessive CPU/GPU or make text difficult to read.
- Reduce particles, shadows and scene detail on mobile or lower-powered devices.
- Suggested tools: Three.js / React Three Fiber for 3D; GSAP and/or Framer Motion for animation. Equivalent tools are acceptable if requirements are met.

9.3 Performance and accessibility

- Core content and actions must remain usable if WebGL is unavailable.
- Provide keyboard navigation and visible focus states.
- Maintain adequate contrast over animated backgrounds.
- Load 3D progressively so core content appears promptly.
- Test homepage and registration on mobile and common browsers.

10. Data Model Overview

Conceptual entities; final columns, constraints and relationships belong in the technical design.

Entity	Purpose / example fields
Users / profiles	Auth ID, name, email, phone, college and required participant details.
Roles / assignments	User, role, department/event scope, granted by and timestamps.
Departments	Name, description and Chief Coordinator assignment.
Events	Department, title, slug, description, category, format, team limits, fee, capacity, deadline, schedule, venue, status and Event Coordinator.
Registrations	Unique ID, event, creator/team leader, source, status and timestamps.
Registration members	Registration ID, participant/member details and role within registration.
Payment proofs	Registration ID, transaction reference, private screenshot path, status, reviewer, reason and timestamps.
Tickets / check-ins	Registration ID, unique token, check-in status, scanner and timestamp.
Announcements	Scope, event/department, title, body, publisher and publish time.
Certificates	Participant or registration, event, private file path, issue status and issuer.
Sponsors / content	Sponsor name, logo, tier, link and editable content blocks.
Audit logs	Actor, action, entity, entity ID, timestamp and limited change metadata.

10.1 Integrity rules

- Use unique IDs for registrations and QR tokens.
- Enforce role scope at database/API level.
- Use safe capacity reservation logic to prevent overbooking.
- Do not store payment screenshots in public storage.
- Avoid unnecessary sensitive data.
- Use transactions or equivalent safeguards for multi-step registration creation.

11. Security and Privacy

- Require authentication for participant and administrative write actions.
- Enforce role-based permissions server-side and through database policies.
- If Supabase is used, apply Row Level Security and never expose service-role secrets in the browser.
- Restrict payment screenshots and certificates to authorised users or intended recipients.
- Validate inputs and restrict upload file types and sizes.
- Rate-limit login, registration and upload endpoints.
- Prevent duplicate submissions and validate capacity safely.
- Audit payment reviews, role changes, event configuration and check-ins.
- Use environment variables for secrets and separate development from production.
- Define a retention/deletion policy for participant data and payment evidence after the fest.
- Prepare backups and a recovery procedure before launch.

12. Analytics and Reporting

Report	Audience	Contents
Registration overview	SAC	Total registrations by department/event and online vs spot.
Event report	Event Coordinator	Registrations, team sizes, remaining capacity and attendance.
Payment review queue	Coordinator / finance	Pending, verified and rejected proof; reference and review timestamps.
Collection summary	Authorised finance / SAC	Expected fees and proof status, clearly separating unverified evidence from verified payments.
Operational readiness	SAC / Chief Coordinator	Events missing venue, schedule, rules, coordinator or publication status.
Export	Authorised users	CSV/Excel download scoped to permitted events or departments.

13. Non-Functional Requirements

Area	Requirement
Performance	Optimise images, fonts, scripts and 3D assets; defer non-critical effects and measure core web vitals.
Scalability	Design for 2,000–5,000 registrations, subject to load testing and selected hosting/database limits.
Responsive design	Support mobile, tablet and desktop; spot registration must work on a phone or laptop at the venue.
Accessibility	Semantic structure, keyboard support, focus indicators, readable contrast and reduced-motion support.
Reliability	Graceful errors, safe retries, duplicate prevention and backups.
Security	Least-privilege access, secure uploads, server-side validation and database policies.
Maintainability	Typed components, reusable UI, documented setup and environment-specific configuration.
Observability	Error logging and basic monitoring for registration, upload and check-in failures.

13.1 Proposed technology stack

Layer	Proposed choice	Notes
Frontend	Next.js + TypeScript	Public site, participant portal and dashboards.
Styling	Tailwind CSS	Responsive UI and design tokens.
3D	Three.js / React Three Fiber	Progressive loading and mobile fallback.
Motion	GSAP and/or Framer Motion	Choose a consistent approach and avoid unnecessary overlap.
Backend/database	Supabase + PostgreSQL	Auth, relational data, storage and row-level security.
Hosting	Vercel	Production deployment and preview environments.

Supabase is the preferred direction in this draft because departments, events, teams, registrations and payment reviews are relational. Confirm the stack before implementation.

14. Acceptance Criteria

- A visitor can browse events and view rules and registration conditions.
- A participant can create a valid individual registration.
- A participant can create a duo/team registration and add required teammates.
- Registration is blocked when the event is closed, full or team rules are violated.
- A paid registration can store screenshot evidence and transaction ID/UTR.
- Submitting proof sets verification pending, not verified.

- An authorised reviewer can verify/reject proof and the decision is auditable.
- An Event Coordinator can create spot registrations for individuals, duos and teams.
- Online and spot registrations appear together with source identified.
- A coordinator cannot access unrelated events unless authorised.
- A QR ticket can be checked in and duplicate check-in is detected.
- Authorised users can export data within their permissions.
- 3D homepage has a fallback and does not block mobile registration.
- Private payment evidence and administrative data are not publicly accessible.

15. Delivery Plan

The target launch window is 1–2 weeks. This is aggressive and assumes timely event content, branding assets, role assignments, fee rules and stakeholder decisions.

Phase	Focus	Deliverables
1. Confirm and design	Scope, content, visual direction, schema and roles	Approved PRD, sitemap, screen designs and event-data template.
2. Public experience	Homepage, 3D hero, event directory and event pages	Responsive public site.
3. Registration core	Authentication, profiles, teams and capacity	Online registration and participant dashboard.
4. Operations	Coordinator dashboards, payment proof and spot registration	Event operations, review queue and exports.
5. Check-in and QA	QR tickets, testing, security and launch	Check-in, acceptance tests, deployment and handover.

If time becomes constrained, prioritise reliable registration and spot registration over extra 3D complexity. Advanced animations and lower-priority dashboards can follow after the core system is stable.

16. Risks and Mitigations

Risk	Impact	Mitigation
Late event information	Incomplete event pages.	Structured event-data template and content freeze date.
3D performance	Slow or unusable mobile experience.	Progressive loading, static fallback, mobile quality tiers and testing.
Capacity race conditions	Overbooking.	Atomic capacity validation/reservation and concurrency tests.
False payment claims	Incorrect paid status or disputes.	Manual verification, audit log and no auto-verification from screenshots.
Spot registration queues	Long entry lines.	Quick mobile-friendly form, minimum required fields, duplicate checks and staff training.
Permission errors	Unauthorised data access.	Database-enforced scope, least privilege and role testing.
Short launch timeline	Core workflows rushed.	Freeze MVP and prioritise operational reliability.
Unconfirmed collection account	Payment operations may not be ready.	Make payment instructions configurable and confirm authorised collection arrangement before launch.

17. Open Decisions and Assumptions

These points were not fully settled and must be confirmed before implementation. They are marked open rather than silently assumed.

Decision	Current understanding	Next step
Accounts for teammates	One person registers and adds teammates; whether each teammate must have an account is unconfirmed.	SAC / product owner decides.
Event publishing	Event Coordinators manage their events; direct publishing vs SAC approval is unconfirmed.	SAC decides.
Payment collection account	UPI/QR, screenshot and transaction ID/UTR with manual review is the intended workflow; account/provider not finalised here.	SAC / authorised finance confirms before launch.
Payment review ownership	Coordinator review was discussed; whether finance must additionally verify is unsettled.	SAC defines roles and audit process.
Ticket eligibility	QR passes are in scope; exact pass activation rule before payment verification is unconfirmed.	SAC decides.
Spot payment handling	Coordinator enters spot registrations; accepted methods and receipt process need confirmation.	SAC / finance defines.
Certificates	Upload/download is desired; eligibility and approval policy need confirmation.	SAC / event coordinators.
Fest dates and content	Dates, schedules, prize details, sponsor assets and contacts were not provided in this PRD.	College supplies approved content.

Post-fest retention	Site is primarily for fest period; retention period for records and payment evidence is unconfirmed.	Institution policy owner decides.

17.1 Confirmed assumptions

- Website serves students from Vardhaman and other colleges.
- Events can be individual, duo or team; each event controls its formats and team sizes.
- Some events are free and some have fees.
- MVP uses UPI/QR instructions and manual payment-proof review, not automatic gateway confirmation.
- Event Coordinators enter spot registrations through the website.
- College manages routine content; technical changes may require developer support.
- Design is dark, futuristic, neon-accented and 3D-led, with usable fallbacks.

Approval checkpoint Before development starts, approve MVP scope, permissions, publishing policy, teammate-account policy, authorised collection arrangement, pass eligibility rules and final event data/content.